

ADCA – Semester 2 Detailed Syllabus

Semester 2 of ADCA generally focuses on **Programming, Web Development, Database Management, Graphics Designing, Accounting (Tally), Networking, and Project Work**. Most institutes include C/C++, HTML, CSS, JavaScript, DBMS, Photoshop, and Tally in this semester.

Subject 1: Programming in C

Unit 1: Introduction to Programming

- What is Programming?
- Algorithm
- Flowchart
- Compiler & Interpreter
- Programming Languages Overview
- Structure of a C Program

Unit 2: C Fundamentals

- Character Set
- Keywords
- Identifiers
- Variables
- Constants
- Data Types
- Operators
 - Arithmetic
 - Relational
 - Logical

- Assignment
- Increment/Decrement

Unit 3: Decision Making

- if Statement
- if-else
- Nested if
- Switch Case
- Conditional Operator

Unit 4: Loops

- while Loop
- do-while Loop
- for Loop
- Nested Loops
- Break & Continue

Unit 5: Arrays & Strings

- One Dimensional Array
- Two Dimensional Array
- String Handling
- String Functions

Unit 6: Functions

- Function Definition
- Function Call
- Return Values
- Recursive Functions

Unit 7: Pointers

- Pointer Basics
- Pointer Arithmetic
- Pointer with Arrays

Unit 8: Structures & File Handling

- Structure Declaration
- Union
- File Opening
- Reading Files
- Writing Files
- Closing Files

Practical Programs

- Addition Calculator
- Simple Interest
- Factorial Program
- Prime Number Check
- Fibonacci Series
- Matrix Addition
- Student Record System
- File Handling Programs

Programming in C commonly covers variables, loops, functions, arrays, pointers, structures, and file handling.

Subject 2: Object-Oriented Programming Using C++

Unit 1: Introduction to OOP

- Procedure-Oriented vs Object-Oriented Programming
- Features of OOP
- Classes and Objects
- Benefits of OOP

Unit 2: Classes and Objects

- Class Definition
- Member Functions
- Access Specifiers
 - Private
 - Public
 - Protected

Unit 3: Constructors & Destructors

- Default Constructor
- Parameterized Constructor
- Destructor

Unit 4: Inheritance

- Single Inheritance
- Multiple Inheritance
- Multilevel Inheritance
- Hierarchical Inheritance

Unit 5: Polymorphism

- Function Overloading
- Operator Overloading
- Virtual Functions

Unit 6: Exception Handling

- try
- catch
- throw

Practical

- Student Class Program
- Employee Management
- Inheritance Examples

- Constructor Programs
- Polymorphism Examples

Many ADCA institutes include C++ with classes, objects, inheritance, polymorphism, constructors, and destructors.

Subject 3: Web Designing (HTML, CSS & JavaScript)

HTML

Introduction

- What is HTML?
- HTML Document Structure
- Tags & Elements

Formatting

- Headings
- Paragraphs
- Text Formatting Tags
- Lists
 - Ordered
 - Unordered
 - Definition

Tables

- Table Creation
- Rowspan
- Colspan

Forms

- Form Elements
- Textbox

- Radio Button
- Checkbox
- Dropdown
- Submit Button

Multimedia

- Images
- Audio
- Video
- Hyperlinks

CSS

Basics

- Inline CSS
- Internal CSS
- External CSS

Styling

- Colors
- Fonts
- Backgrounds
- Borders
- Margin
- Padding

Layout

- Flexbox Basics
- Grid Basics
- Responsive Design

JavaScript

Basics

- Variables

- Data Types
- Operators

Control Statements

- if
- switch
- loops

Functions

- User Defined Functions
- Arrow Functions

DOM Manipulation

- getElementById()
- querySelector()
- Events

Validation

- Form Validation
- Input Checking

Practical

- Personal Website
- Registration Form
- Login Page
- Portfolio Website
- Calculator Project

Web Designing generally includes HTML, CSS, JavaScript, forms, tables, layouts, and event handling.

Subject 4: Database Management System (DBMS)

Unit 1: Introduction to DBMS

- Database Concepts
- Advantages of DBMS
- Data Models

Unit 2: RDBMS

- Tables
- Rows
- Columns
- Keys
 - Primary Key
 - Foreign Key

Unit 3: SQL

DDL Commands

- CREATE
- ALTER
- DROP

DML Commands

- INSERT
- UPDATE
- DELETE

DQL Commands

- SELECT
- WHERE
- ORDER BY
- GROUP BY

Unit 4: Joins

- Inner Join
- Left Join

- Right Join
- Full Join

Unit 5: Normalization

- 1NF
- 2NF
- 3NF

Practical

- Student Database
- Employee Database
- Inventory Database
- SQL Queries
- Report Generation

DBMS modules typically include tables, keys, SQL commands, joins, and normalization.

Subject 5: Tally Prime / Tally ERP with GST

Unit 1: Accounting Fundamentals

- Accounting Concepts
- Debit & Credit Rules
- Journal
- Ledger
- Trial Balance

Unit 2: Company Management

- Company Creation
- Company Alteration
- Company Backup

Unit 3: Ledger & Groups

- Ledger Creation
- Group Creation

Unit 4: Voucher Entry

- Payment Voucher
- Receipt Voucher
- Contra Voucher
- Journal Voucher
- Purchase Voucher
- Sales Voucher

Unit 5: Inventory Management

- Stock Group
- Stock Category
- Units of Measurement
- Godown

Unit 6: GST

- GST Setup
- Purchase Entry
- Sales Entry
- GST Reports

Unit 7: Reports

- Trial Balance
- Profit & Loss Account
- Balance Sheet
- Day Book

Practical

- Create Company
- GST Billing
- Inventory Records

- Financial Statements

Tally modules usually cover company creation, ledgers, vouchers, inventory, GST, and final accounts.

Subject 6: Photoshop & Graphic Designing

Unit 1: Introduction to Photoshop

- Interface
- Workspace
- Tools Panel

Unit 2: Selection Tools

- Marquee Tool
- Lasso Tool
- Magic Wand

Unit 3: Layers

- Layer Creation
- Layer Management
- Layer Effects

Unit 4: Image Editing

- Crop
- Resize
- Retouching
- Background Removal

Unit 5: Color Correction

- Brightness
- Contrast
- Hue/Saturation

Unit 6: Design Projects

- Banner Design
- Poster Design
- Visiting Card
- Social Media Post
- Logo Design

Practical

- Passport Photo Editing
- Background Change
- Poster Creation
- Advertisement Design
- Wedding Album Design

Graphics modules commonly include Photoshop, image editing, layers, effects, and poster/banner design.

Subject 7: Computer Networking & Cyber Security

Networking

- Network Fundamentals
- LAN
- MAN
- WAN
- Network Topologies
- Router
- Switch
- Hub
- Modem
- IP Address

- TCP/IP

Cyber Security

- Malware
- Virus
- Worm
- Trojan
- Ransomware
- Phishing
- Password Security
- Antivirus
- Firewall
- Safe Internet Practices

Networking and cyber security basics are commonly included in advanced ADCA programs.

Subject 8: Project Work

Students generally prepare one major project using any of the following technologies:

Web Project

- College Website
- School Website
- Portfolio Website
- Hospital Website

C/C++ Project

- Library Management System
- Student Management System

- Banking System

Database Project

- Inventory Management
- Employee Database

Tally Project

- GST Billing System
- Company Accounts Management

Project work is a standard component of Semester 2.

Viva Questions Preparation

1. What is DBMS?
 2. Difference between C and C++?
 3. What is a Class?
 4. What is an Object?
 5. What is HTML?
 6. Difference between HTML and CSS?
 7. What is JavaScript?
 8. What is Primary Key?
 9. What is GST?
 10. What is a Ledger?
 11. What is a Voucher?
 12. What is LAN?
 13. What is IP Address?
 14. What is a Virus?
 15. What is a Firewall?
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Skills After Completing ADCA

After Semester 2, you should be able to:

- Build websites using HTML, CSS, and JavaScript
- Write C and C++ programs
- Design databases using SQL
- Manage accounts in Tally with GST
- Create posters and graphics in Photoshop
- Understand networking and cybersecurity fundamentals
- Develop small software and web projects.